

Steps for Deployment

Following are the steps for deployment :

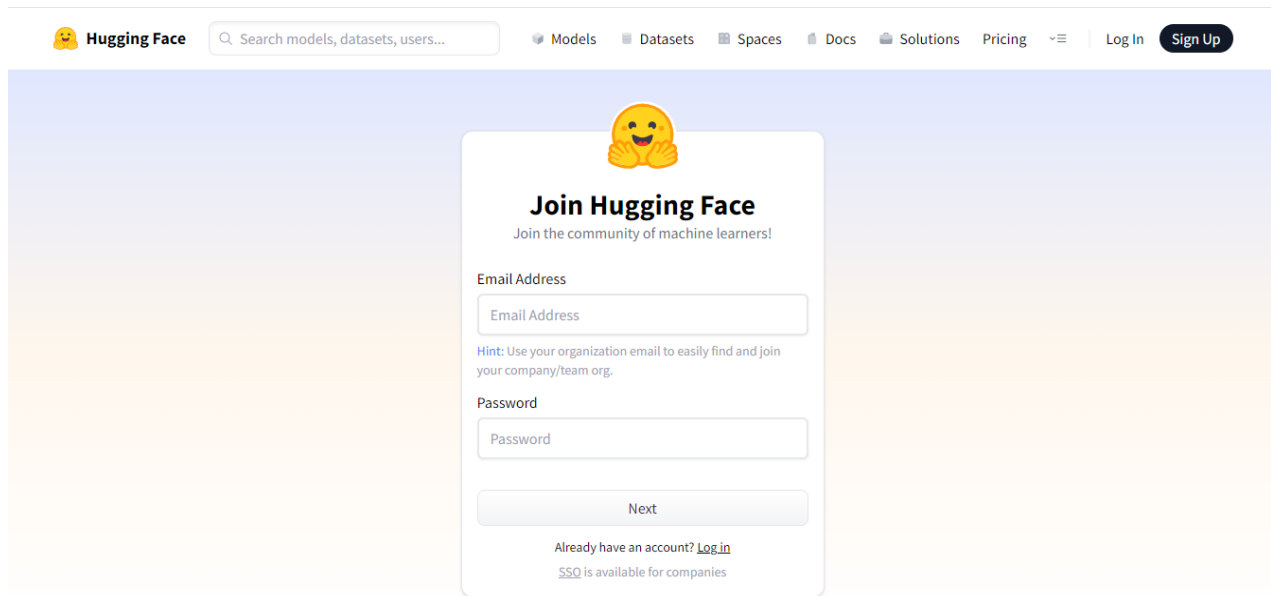
A. Hackathon3a_Face_Similarity_and_Recognition

Stage 1: Build a Siamese Network and obtain the Siamese Representation for the AT&T faces dataset

Once you have done with Stage 1
Proceed further with steps below in Stage 2:

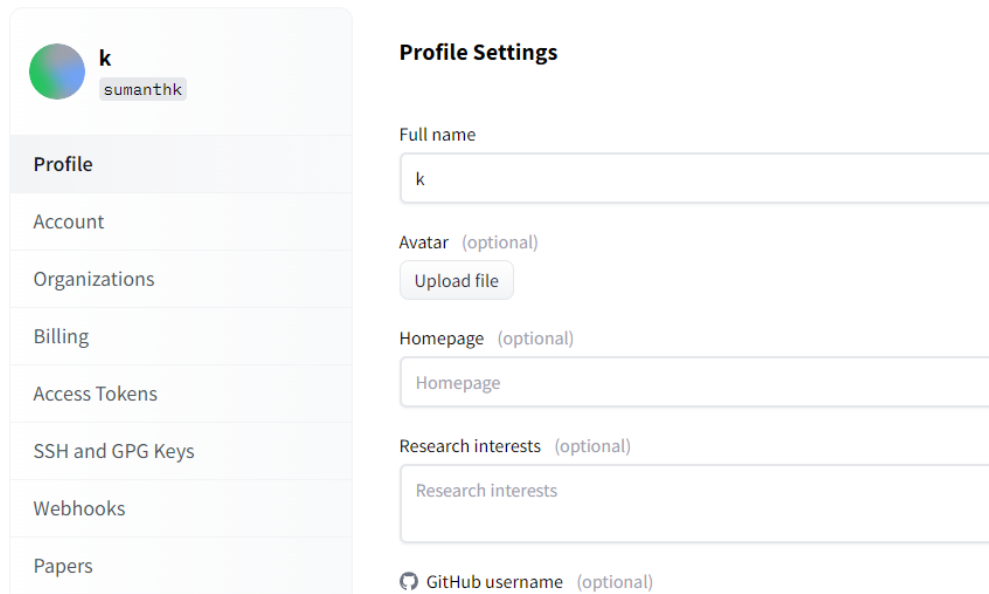
Stage 2 :

1. [Sign up](https://huggingface.co/join) for a Hugging Face account



2. Once you have an account, create an access token for your account and save it.

→ Go to your 'Settings' → under your profile icon, then click on the → 'Access Tokens' tab



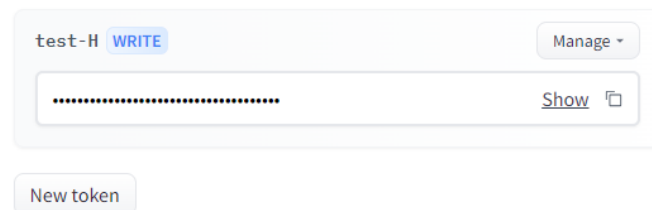
The image shows the GitHub 'Profile Settings' page. On the left is a sidebar with navigation links: Profile, Account, Organizations, Billing, Access Tokens, SSH and GPG Keys, Webhooks, and Papers. The 'Profile' link is selected. The main content area is titled 'Profile Settings' and contains several form fields: 'Full name' (with the value 'k'), 'Avatar (optional)' with an 'Upload file' button, 'Homepage (optional)' (with the value 'Homepage'), 'Research interests (optional)' (with the value 'Research interests'), and 'GitHub username (optional)'.

→ Click on the 'New token' button to create a new User Access Token.

Access Tokens

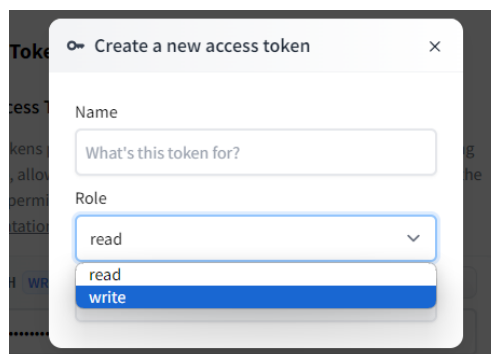
User Access Tokens

Access tokens programmatically authenticate your identity to the Hugging Face Hub, allowing applications to perform specific actions specified by the scope of permissions (read, write, or admin) granted. Visit [the documentation](#) to discover how to use them.



The image shows a list of user access tokens. One token is visible with the name 'test-H' and the role 'WRITE'. To the right of the token name is a 'Manage' button with a dropdown arrow. Below the token name is a field containing a masked token (dots) and a 'Show' button with a copy icon. At the bottom of the list is a 'New token' button.

→ Select a role('write') and a name for your token

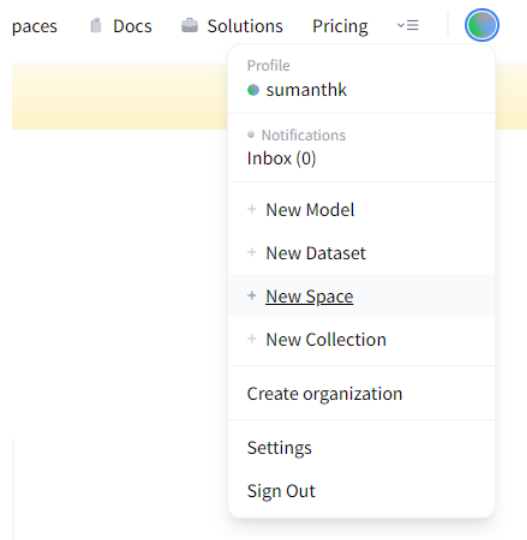


The image shows a modal dialog titled 'Create a new access token'. It has two main fields: 'Name' with a placeholder 'What's this token for?' and 'Role' which is a dropdown menu. The dropdown menu is open, showing two options: 'read' and 'write', with 'write' selected and highlighted in blue.

→ Click Generate a token

To know more about Access Tokens, refer
[here](https://huggingface.co/docs/hub/security-tokens).

3. Start a new Hugging Face Space by going to your profile and clicking → "New Space"
[https://huggingface.co/new-space]



4. Provide details for your space:

- Space name
- License (eg. [MIT])(<https://opensource.org/licenses/MIT>))
- For Space SDK (software development kit), select 'Docker' (**It is mandatory to select *Docker* for this application to work.**)
- Choose a Docker template: 'Blank'



Create a new Space

Spaces are Git repositories that host application code for Machine Learning demos.
You can build Spaces with Python libraries like Streamlit or Gradio, or using Docker images.

Owner

sumanthk

Space name

New space name

License

License

Select the Space SDK

You can choose between Streamlit, Gradio and Static for your Space. Or pick Docker to host any other app.


Streamlit


Gradio


Docker
10 templates


Static

Choose a Docker template:

 Blank

 Argilla

 Livebook

 LabelStudio

- Space hardware (CPU basic)
- Choose whether your Space is public or private
- Click "Create Space"
- After Space creation, the files of your repo can be seen via `Files` tab

Space hardware Free

CPU basic · 2 vCPU · 16 GB · FREE

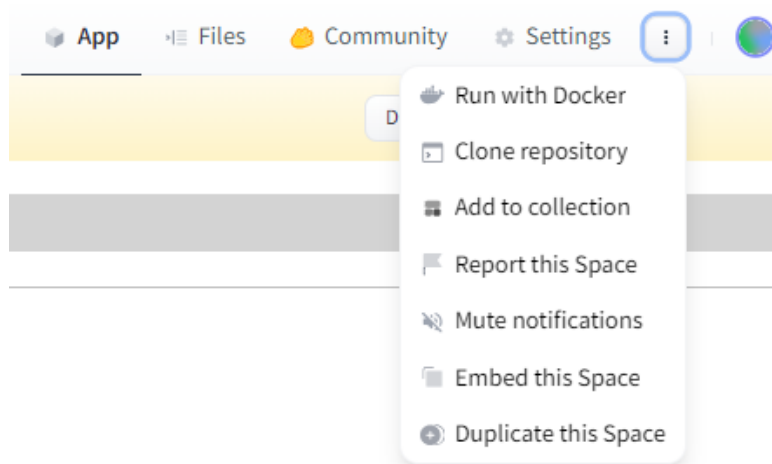
You can switch to a different hardware at any time in your Space settings.
You will be billed for every minute of uptime on a paid hardware.

- ☒  **Public**
Anyone on the internet can see this space. Only you (personal space) or members of your organization (organization space) can commit.
- ☐  **Private**
Only you (personal space) or members of your organization (organization space) can see and commit to this space.

Create Space

5. Clone your space repository on to your system

- Go to 3-dots icon beside Settings, then select `Clone Repository` option
- `git lfs install`
- `git clone`
- Run clone commands in your terminal (Provide your HuggingFace username and access token password if prompted for authentication.)



6. Download the project folder `recognition\_api`, shared in drive along with this notebook, on to your system.

7. Add below files from `recognition\_api` folder to your cloned repository folder

- `app`, `requirements.txt`, `Dockerfile`, and `.gitattributes`
- You may need to replace the `.gitattributes` file if it is already present in repository
- NOTE: Do NOT commit and push changes yet.

Name	Owner	Last modified	File size
app	Yograj Mahawar	Oct 24, 2023	—
.gitattributes	Yograj Mahawar	Oct 23, 2023	2 KB
Dockerfile	Yograj Mahawar	Oct 24, 2023	538 bytes
requirements.txt	Yograj Mahawar	Oct 23, 2023	284 bytes

8. Update files present in your cloned repository

- Download the Siamese model(trained in colab for face similarity) and place it within `app/Hackathon\_setup/` folder
- Update the Siamese model architecture in the `app/Hackathon\_setup/face\_recognition\_model.py` file and provide the code in the `get\_similarity()` function of the `app/Hackathon\_setup/face\_recognition.py` file. (See Deployment related files)
- Update the `\*\*README.md\*\*` file and include `app\_port: 8001`

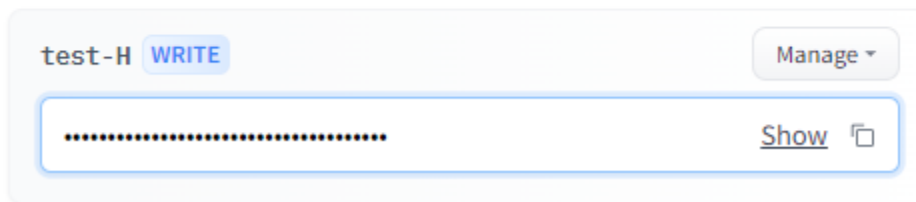
```
1 ---
2 title: Hackfour
3 emoji: 🐱
4 colorFrom: yellow
5 colorTo: pink
6 sdk: docker
7 pinned: false
8 license: mit
9 app_port: 8001
10 ---
11
12 Check out the configuration reference at https://github.com/robertocarreras/hack4u
13
```

9. Commit your changes and push to HuggingFace Space repository(Provide your HuggingFace username and for password go to access token copy the prompt for authentication.)

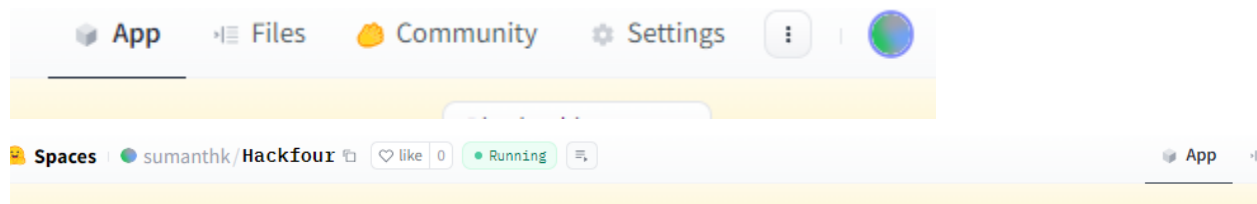
Access Tokens

User Access Tokens

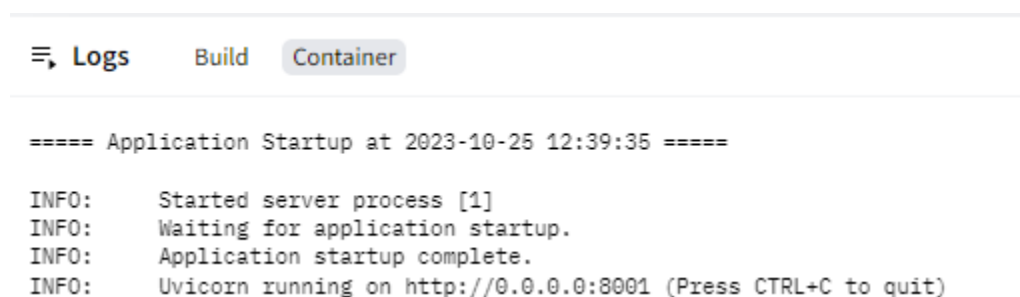
Access tokens programmatically authenticate your identity to the Hugging Face Hub, allowing applications to perform specific actions specified by the scope of permissions (read, write, or admin) granted. Visit [the documentation](#) to discover how to use them.



10. Access the `App` tab of your repository to see the build progress (debug if error persists, check logs using `App` tab and `logs` icon beside running icon) [This step may take some time.]

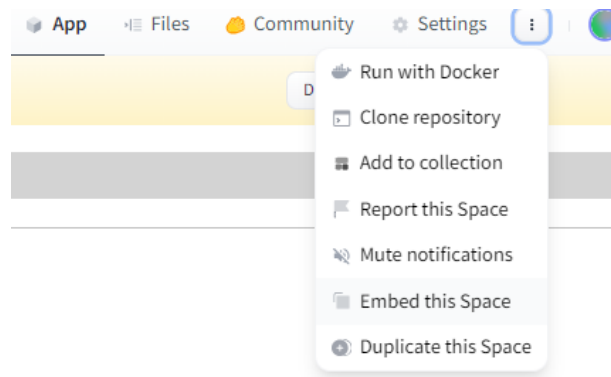


11. Once the app has built successfully, you should see below message
`Application startup complete. Uvicorn running on http://0.0.0.0:8001`



12. Test the Siamese model using the application running on your Space

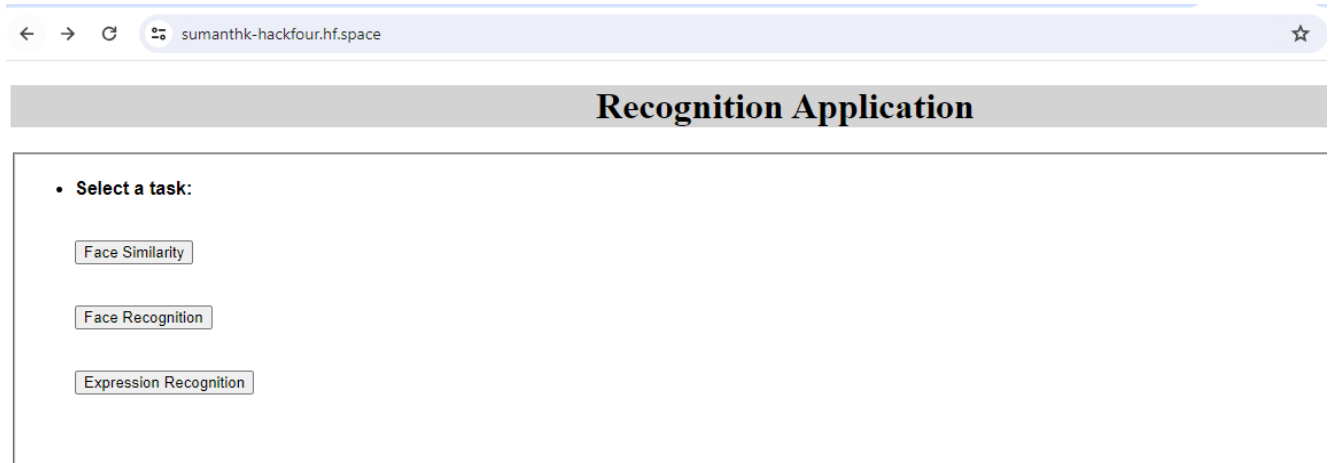
- Go to 3-dots icon beside Settings, then select `Embed this Space` option, and copy `Direct URL` and search in browser



Note: In a case if embed space did not showup, in this case you can create a URL directly in this way : <https://XXusername-XXprojectname.hf.space/> .
from the below image “**sumanthk**” -> is username and “**hackfour**” -> is project name.



This html page will open up.



The screenshot shows a web browser window with the address bar displaying 'sumanthk-hackfour.hf.space'. The page title is 'Recognition Application'. Below the title, there is a section titled 'Select a task:' with three radio button options: 'Face Similarity', 'Face Recognition', and 'Expression Recognition'. The 'Face Similarity' option is selected.

- Select the task, 'Face Similarity' for now

- Select 'Send Anyway' when prompted



The information you're about to submit is not secure

Because this form is being submitted using a connection that's not secure, your information will be visible to others.

Send anyway

Go back

- Upload images and test

- NOTE: When using the Direct URL link via mobile, the camera option will also enable to capture images (Set 1:1 aspect ratio in camera settings before-hand)

B. Hackathon3b_Expression_Recognition

(i) Train a CNN Model for Expression Recognition on given Expression data

(ii) Deploy the Model and Perform Expression Recognition on Team Data through the HuggingFace Space App

Repeat the procedure similar to the earlier Hackathon3a_Face_Similarity_and_Recognition for deployment

- ❖ Define and train a CNN for expression recognition for the data under folder `"Expression_data"` which is segregated on expression basis.
- ❖ Collect your team data by running the code cells provided within the notebook below.
- ❖ Test your model on the collected team data and optimize the CNN architecture for predicting the respective labels of the images.
- ❖ Update files present in your cloned HuggingFace Space repository.
- ❖ Save and Download the trained expression model (``expression_model.t7``) and upload/place it in your HuggingFace Space repository within the ``app/Hackathon_setup/`` folder.
- ❖ Update the model architecture in the ``app/Hackathon_setup/exp_recognition_model.py`` file.
- ❖ Update the code in the ``get_expression()`` function of the ``app/Hackathon_setup/exp_recognition.py`` file. (See Deployment related files)
- ❖ Commit your changes and push to the HuggingFace Space repository.
- ❖ Access the ``App`` tab of your repository to see the build progress (debug if error persists) [This step may take some time.]
- ❖ Once the app has built successfully, you should see below message ``Application startup complete. Uvicorn running on http://0.0.0.0:8001``
- ❖ Test the model's Expression Recognition functionality using the application running on your Space. Go to 3-dots icon beside Settings, then select ``Embed this Space`` option, and go to ``Direct URL``
- ❖ Select the task, ``Expression Recognition`` → Select ``Send Anyway`` when prompted → Upload your image and test

NOTE: When using the Direct URL link via android mobile, the camera option will also enable to capture images (Set 1:1 aspect ratio in camera settings before-hand)